(Questions)

Fertilizer (X)	yield	XY	X^2
100	40	4000	10,000
200	50	10,000	40,000
300	50	15,000	90,000
400	70	28,000	16,0000
500	65	32,500	250,000
600	65	39,000	360,000
700	80	56,000	49,0000
$\Sigma X = 2800$	$\Sigma Y = 420$	$= \Sigma XY$	$\Sigma X^2 =$
		184500	140,0000

$$\therefore \beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\therefore \beta_1 = \frac{n\Sigma XY - \Sigma X \Sigma Y}{n\Sigma X^2 - (\Sigma X)^2}$$

$$\beta_1 = \frac{(7)(184500) - (2800)(420)}{(7)(140,0000) - (2800)^2}$$

$$= \frac{1291500 - 1176000}{980,0000 - 784,0000}$$

$$= \frac{115500}{196,0000} \Rightarrow 0.0589$$

$$\bar{Y} = \frac{\Sigma Y}{n} = \frac{420}{7} = 60$$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{2800}{7} = 400$$

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$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\beta_0 = 60 - (0.0589)(400)$$

$$= 60 - 235.6$$

$$\boxed{\beta_0 = -175.6}$$

$$\hat{Y} = -175.6 - 0.0589X.$$

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#2:-

X	Y	XY	X^2
15	60	900	225
24	45	1080	576
25	50	1250	625
30	35	1050	900
35	42	1470	1225
40	46	1840	1600
45	28	1260	2025
65	20	1300	4225
70	22	1540	4900
75	15	1125	5625
$\Sigma X = 424$	$\Sigma Y = 363$	$\Sigma XY = 12815$	$\Sigma X^2 = 21926$

$n = 10$

$\therefore \beta_0 = \bar{Y} - \beta_1 \bar{X}$

$\therefore \beta_1 = \dfrac{n\Sigma XY - \Sigma X \Sigma Y}{n\Sigma X^2 - (\Sigma X)^2}$

$\beta_1 = \dfrac{(10)12815 - (424)(363)}{(10)21926 - (424)^2}$

$= \dfrac{128150 - 153912}{219260 - 179776}$

$= \dfrac{-25762}{\cancel{95734}\ 39484} \Rightarrow \cancel{-0.9691}$

$\Rightarrow -0.6524$

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$$\bar{Y} = \frac{\Sigma Y}{n} = \frac{363}{10} \Rightarrow 36.3$$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{424}{10} \Rightarrow 42.4$$

$$\hat{\beta}_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\hat{\beta}_0 = 36.3 - (-0.6524)(42.4)$$

$$= 36.3 + 27.06645$$

$$\hat{\beta}_0 = 63.796$$

$$\hat{Y} = \beta_0 - \beta_1 X$$

$$\hat{Y} = 63.96 - (-0.6524) X$$

$$\hat{Y} = 63.796 + 0.6594 X$$

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#03:-

X Weight	Y Chest.S	XY	X^2
2.71	29.5	79.945	7.3441
2.15	26.3	56.545	4.6225
4.40	32.2	141.68	19.36
5.50	36.5	200.75	30.25
3.20	27.2	87.04	10.24
4.30	27.5	118.25	18.49
2.31	28.3	65.373	5.3361
4.30	30.3	130.29	18.49
3.70	28.5	105.45	13.69

$\Sigma X = 32.57$ $\Sigma Y = 266.3$ $\Sigma XY = 985.323$ $\Sigma X^2 = 127.8227$

$$\beta_1 = \frac{(9)(985.323) - (32.57)(266.3)}{(9)(127.8227) - (32.57)^2}$$

$$= \frac{8867.907 - 8673.391}{1150.4043 - 1060.8049}$$

$$= \frac{194.516}{289.5994}$$

$$\boxed{\beta_1 = 0.6716}$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\bar{Y} = \frac{266.3}{9} \Rightarrow 29.59$$

$$\bar{X} = \frac{32.57}{9} \Rightarrow 3.619$$

$$\beta_0 = 29.59 - (0.6716)(3.619) \Rightarrow 27.15$$

29.59 − 2.4 3[illegible]

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Day:

$$\hat{Y} = \beta_0 - \beta_1 X$$

$$\hat{Y} = 26.28 - 0.9140X$$

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#04 :-

X Paper I	Y Paper II	XY	X^2
45	56	2520	2025
55	50	2750	3025
56	48	2688	3136
58	60	3480	3364
60	62	3720	3600
65	64	4160	4225
68	65	4420	4624
70	70	4900	4900
75	74	5550	5625
80	82	6560	6400
85	90	7650	7225
$\Sigma X = 717$	$\Sigma Y = 721$	$\Sigma XY = 48,398$	$\Sigma X^2 = 48,149$

$$\beta_1 = \frac{(11)(48,398) - (717)(721)}{(11)(48,149) - (717)^2}$$

$$= \frac{532378 - 516957}{529639 - 514089}$$

$$= \frac{15421}{15550} \Rightarrow 0.9917$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\bar{Y} = \frac{721}{11} \Rightarrow 65.545$$

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$$\bar{X} = \frac{717}{11} \Rightarrow 65.18$$

$$\beta_0 = 65.545 - (0.9917)(65.18)$$

$$\beta_0 = 65.545 - 64.641$$

$$\beta_0 = 0.904$$

$$\hat{Y} = \beta_0 - \beta_1 X$$

$$\hat{Y} = 0.904 - 0.9917X$$

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